Calculation of current-voltage characteristic of a resonant-tunneling diode (RTD) and using the adiabatic approximation for simulation of conductance quantization in quantum point contact (QPC)

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Contents

1 Objective **2**

2 Results **2**

2.1 Transfer Matrix Method - Single Barrier 2

2.2 Resonant-Tunneling Diode - Double Barrier 2

2.3 Current-Voltage Characteristic of the RTD 3

2.4 Quantum Point Contact - Adiabatic Approximation 4

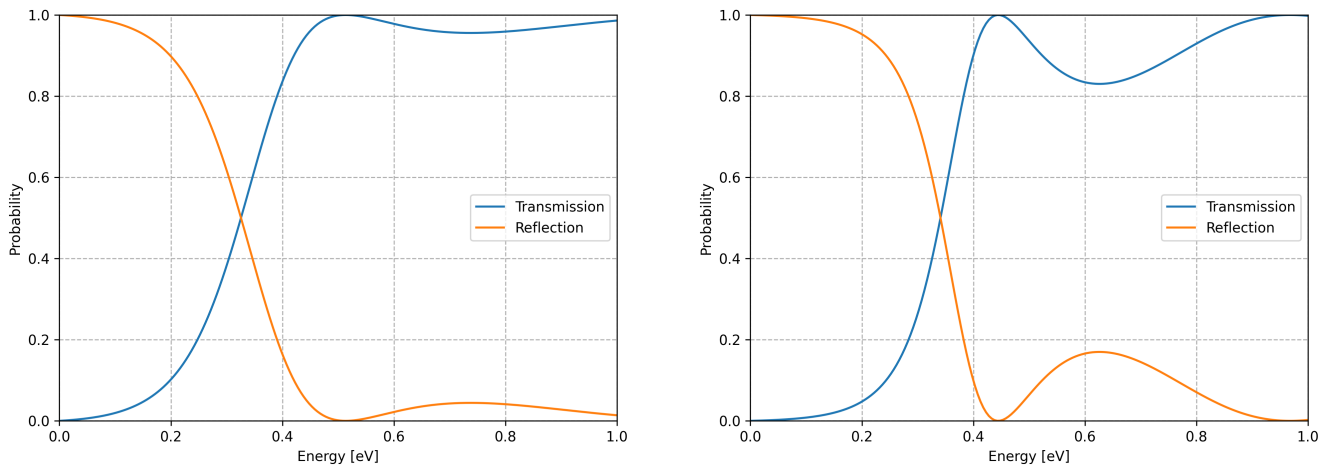
1 Objective

The aim of this exercise was to calculate transmission and reflection coefficients for semiconductor nanostructures using the transfer matrix (monodromy matrix) method, to compute the current-voltage characteristic of a resonant-tunneling diode (RTD) via the Tsu-Esaki formula, and to simulate conductance quantization in a quantum point contact (QPC) within the adiabatic approximation.

2 Results

2.1 Transfer Matrix Method - Single Barrier

A GaAs/$Al_{0.3}Ga_{0.7}As$ structure with a single potential barrier of width $d_{\mathrm{AlGaAs}} = 5$ nm and height $U = 0.27$ eV was considered. The effective masses were taken as $m^*_{\mathrm{GaAs}} = 0.063\, m_0$ and $m^*_{\mathrm{AlGaAs}} = (0.063 + 0.083x)\, m_0$ for $x = 0.3$.



(a) Constant effective mass $m^* = m^*_{\mathrm{GaAs}}$.

(b) Varying effective mass in GaAs and AlGaAs.

Figure 1: Transmission $T(E)$ and reflection $R(E)$ coefficients for a single potential barrier.

In both cases, transmission increases monotonically with energy. Above the barrier height ($E > 0.27$ eV) the electron passes classically, leading to a rapid rise in $T(E)$. Below this energy, the electron tunnels through the barrier and transmission is exponentially suppressed. Including a spatially varying effective mass slightly reduces the sub-barrier transmission compared to the constant-mass case.

2.2 Resonant-Tunneling Diode - Double Barrier

A structure consisting of two AlGaAs barriers (each of width $d_{\mathrm{AlGaAs}} = 5$ nm) separated by a GaAs quantum well of width $d_{\mathrm{GaAs}} = 3$ nm was studied.

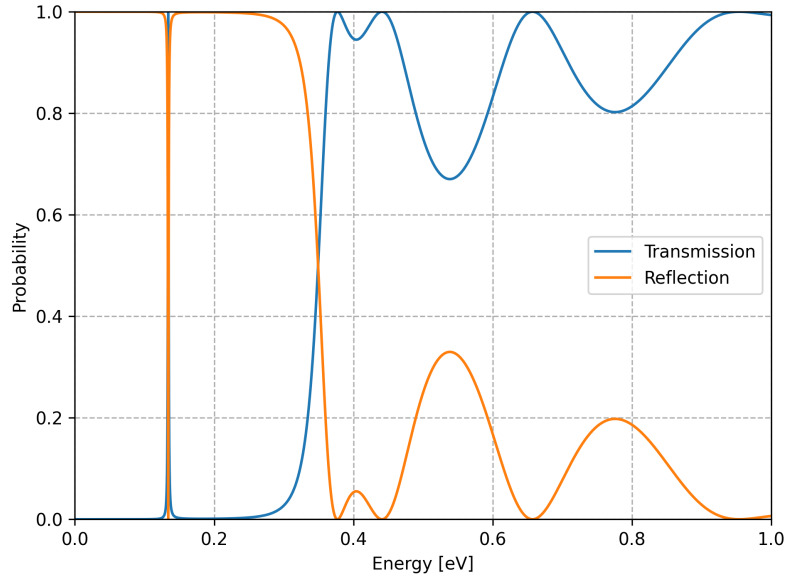


Figure 2: Transmission and reflection coefficients for the double-barrier RTD structure. A resonant transmission peak is clearly visible.

A sharp resonant peak is observed in the transmission spectrum at the energy corresponding to the bound state of the quantum well. This is resonant tunneling. When the electron energy matches the quasi-bound level of the well, transmission reaches $T = 1$ despite the presence of two potential barriers.

2.3 Current-Voltage Characteristic of the RTD

The current density $j(V_{\text{bias}})$ was calculated using the Tsu-Esaki formula:

$$j = \frac{em^* k_B T}{2\pi^2 \hbar^2} \int_0^\infty dE_z \, T(E_z) \ln \left[\frac{1 + e^{(\mu_s - E_z)/k_B T}}{1 + e^{(\mu_d - eV_{\text{bias}} - E_z)/k_B T}} \right], \tag{1}$$

with parameters $\mu_s = \mu_d = 87$ meV and $T = 77$ K. The applied bias was modelled as a linear potential drop along the nanostructure.

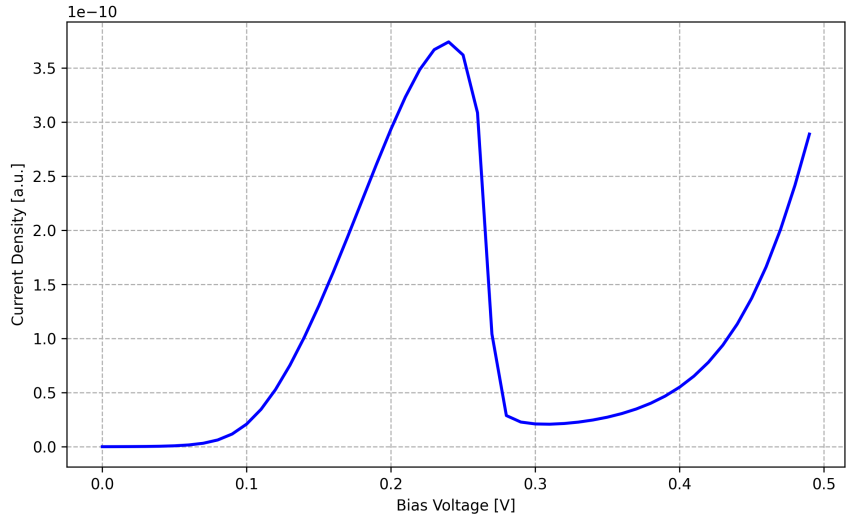


Figure 3: Current-voltage characteristic of the RTD computed with the Tsu-Esaki formula.

The characteristic displays a clear region of negative differential resistance (NDR), where the current decreases with increasing voltage. This occurs when, at higher bias, the resonant level of the well shifts out of the energy window defined by the chemical potentials, causing the transmission to drop sharply.

2.4 Quantum Point Contact - Adiabatic Approximation

The QPC was defined in a two-dimensional electron gas (2DEG) with dimensions $W = 50$ nm and $L = 100$ nm, with gates spanning $[0.3, 0.7]L$. The gate voltage was $V_g = 4$ V, dielectric constant $\varepsilon = 13.6$, and gate-to-2DEG distance $d = 3$ nm.

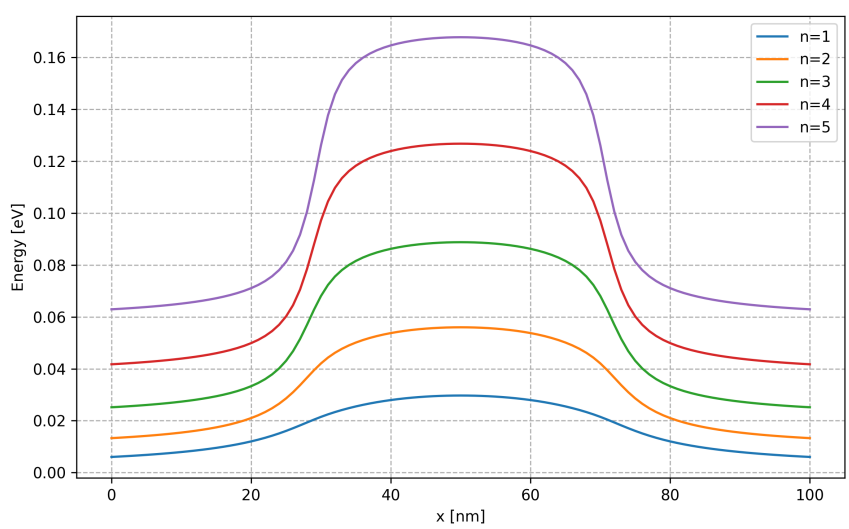
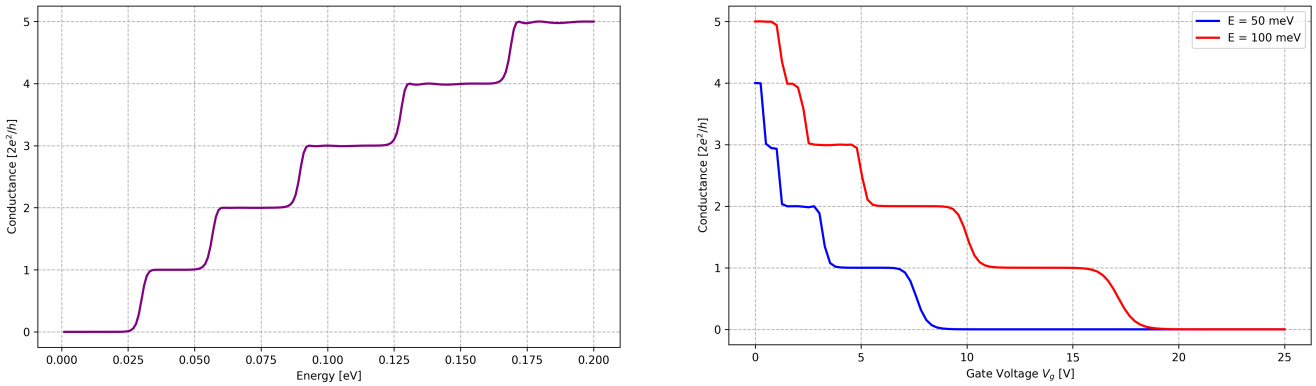


Figure 4: Effective potential $E_n(x)$ for the five lowest transverse modes of the QPC at $V_g = 4$ V.

The effective potential $E_n(x)$ forms a barrier at the center of the device, with higher transverse modes experiencing progressively higher barriers. Conductance was computed via the Landauer

formula:

$$G = \frac{2e^2}{h} \sum_n T_n(E). \tag{2}$$



(a) Conductance $G(E)$ as a function of energy.

(b) Conductance $G(V_g)$ for $E = 50$ and 100 meV.

Figure 5: Conductance of the quantum point contact.

The $G(E)$ plot shows stepwise increases of $2e^2/h$ at threshold energies corresponding to successive transverse modes opening, which means the conductance is quantized. In the $G(V_g)$ plot, analogous steps appear as a function of gate voltage: as V_g increases, the conductance drops in discrete decrements.